

An Overview of Algebraic Multigrid

Computational Math Seminar, October 2022

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Let $\mathbf{A}\mathbf{u} = \mathbf{f}$ be a sparse linear system arising from the discretization of an elliptic PDE in the domain Ω . (Last time we looked at $\nabla^2 u = f$ in 1-D)

Notation:

- $\mathbf{v}$ is a current guess or approximation to the solution
- $\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{v}$ is the residual
- $\mathbf{e} = \mathbf{u} - \mathbf{v}$ is the error in the current guess
- $\mathbf{A}\mathbf{e} = \mathbf{r}$ is the error equation
- Ω^h is the mesh with grid spacing h

Multigrid Algorithm

Two-grid Correction Scheme:

Given $\mathbf{A}^h, \mathbf{f}^h$:

- 1 Relax on $\mathbf{A}^h \mathbf{v}^h = \mathbf{f}^h$ on Ω^h
- 2 Compute $\mathbf{r} = \mathbf{f}^h - \mathbf{A}^h \mathbf{v}^h$
- 3 Restrict $\mathbf{A}^h, \mathbf{r}^h$ to Ω^{2h}
- 4 Solve $\mathbf{A}^{2h} \mathbf{e}^{2h} = \mathbf{r}^{2h}$
- 5 Interpolate $\mathbf{e}^{2h}$ to Ω^h and let $\mathbf{v}^h \leftarrow \mathbf{v}^h + \mathbf{e}^h$
- 6 Relax on $\mathbf{A}^h \mathbf{v}^h = \mathbf{f}^h$ on Ω^h

Complexity: $O(N)$ where N is the number of unknowns

Relaxation/Smoothing

- Goal is to approximately solve $\mathbf{A}\mathbf{v} = \mathbf{f}$ using a cheap iterative procedure
- Can be written in the form $\mathbf{v} \leftarrow \mathbf{v} + \mathbf{B}\mathbf{r}$
- Can also be written as a fixed point iteration applied the error:
 $\mathbf{e} \leftarrow \mathbf{R}\mathbf{e}$
- $\mathbf{B}$, $\mathbf{R}$ make *local* corrections
- Examples: Jacobi, Gauss-Seidel, SOR

Spectral Perspective on Relaxation

- Think about Ω^h as vector space $\mathbb{R}^N$ where N is the number of unknowns on the fine grid
- Can decompose the error into Fourier modes on the grid
- Relaxation methods typically are effective at reducing oscillatory modes but are ineffective at reducing smooth modes
- Geometrically smooth modes are associated with small residuals - think about null space of ∇^2 (zero row sum)

Spectral Perspective on Restriction/Interpolation

- Smooth error on fine grid is oscillatory on coarse grid
- Smooth error on coarse grid can accurately be interpolated onto fine grid
- Recursive application of two-grid correction scheme eliminates all modes

Multigrid Summary

Elements of Multigrid:

- 1 Relaxation Method
- 2 Sequence of Coarse Grids
- 3 Restriction Operator
- 4 Interpolation Operator
- 5 Coarse-Grid Operator

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Algebraic Perspective on Relaxation

- Have the orthogonal decomposition $\Omega^h = L \oplus H$ where L represents modes with wavenumber $1 \leq k < n/2$ and H represents modes with wavenumber $n/2 \leq k \leq n$
- Relaxation on fine grid eliminates component of error in $\mathbf{H}$
- Smooth errors are associated with small residuals - think about constants
- Smooth errors are associated with small eigenvalues - think about nearing the null space

Algebraic Perspective on Restriction/Interpolation

- Let $\mathbf{I}_h^{2h}$ be the restriction operator and $\mathbf{I}_{2h}^h$ be the interpolation operator
- For model problem we have $\mathbf{A}^{2h} = \mathbf{I}_h^{2h} \mathbf{A}^h \mathbf{I}_{2h}^h$ and $\mathbf{I}_h^{2h} = c(\mathbf{I}_{2h}^h)^T$
- Two-grid Correction scheme without relaxation:
 $\mathbf{TG} = \mathbf{I} - \mathbf{I}_h^{2h}(\mathbf{A}^{2h})^{-1} \mathbf{I}_h^{2h} \mathbf{A}^h$
- We have subspace decomposition $\Omega^h = \mathcal{R}(\mathbf{I}_{2h}^h) \oplus \mathcal{N}(\mathbf{I}_h^{2h})$
- Can also show $\mathcal{R}(\mathbf{I}_{2h}^h) = \mathcal{N}(\mathbf{TG})$

Algebraic Perspective on Two-Grid correction Scheme

- Relaxation on fine grid eliminates component of error in $\mathbf{H}$
- Two-grid correction scheme eliminates component of error in $\mathcal{R}(\mathbf{I}_{2h}^h) = \mathcal{N}(\mathbf{TG})$
- Think about this as fixed point iteration driving $\mathbf{e}$ toward the origin
- Coarse grid correction should eliminate components of error in range of interpolation

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Towards Algebraic Multigrid

Now suppose we have a sparse matrix $\mathbf{A}$ which arises from the discretization of a complicated second-order elliptic PDE. How can we define a multigrid scheme based on just the entries of $\mathbf{A}$ and $\mathbf{f}$?

- Recall relaxation stalls after eliminating component of error in high frequency range
- Therefore, DEFINE smooth error as that error not eliminated by relaxation
- Need a coarse grid that can accurately represent this smooth error
- Need an interpolation operator that can accurately transfer smooth error

Assume this $\mathbf{A}$ is an s.p.d. M-matrix - positive diagonal, non-positive off diagonal

Interpolation Operator

Consider indices $i = 1, 2, \dots, u$ to be “grid” points. A coarse grid is some subset of these indices. Suppose we have already divided these into two sets $C \subset F$, where F is the set of fine grid points and C is the coarse grid points.

Defintion: u_j *strongly influences* u_i if $|a_{ij}| \geq \theta \max_{k \neq i} |a_{ik}|$

- Can be shown that a coarse grid point j strongly influences a fine grid point i then $e_j \approx e_i$ - smooth error varies slowly in the direction of strong connection
- Build interpolation operator using assumption that $\mathbf{Ae} \approx 0$ for smooth error

Coarse Grid Selection

Guided by two heuristics:

- ① Grid points that strongly influence i should be in the coarse set or at least strongly depend on point(s) in the coarse set
 - ② Set of coarse points should be maximal in terms of subsets that have the property that none of the coarse points depend strongly on each other
- Coarse-grid correction proceeds by two-pass algorithm: first use heuristics to create initial partition into coarse and fine, then go back and enforce heuristic 1
 - Coarsens in the direction of strong connection, because smooth error varies slowly in direction of strong connection

Algebraic Multigrid Algorithm

- 1 Build sequence of grids using two pass algorithm
- 2 Build interpolation operators based on concept of strong influence
- 3 Let restriction operator be transpose of interpolation
- 4 Define $\mathbf{A}^{2h} = \mathbf{I}_h^{2h} \mathbf{A}^h \mathbf{I}_{2h}^h$ - this leads to a coarse grid correction that is optimal in $\mathbf{A}^h$ -norm